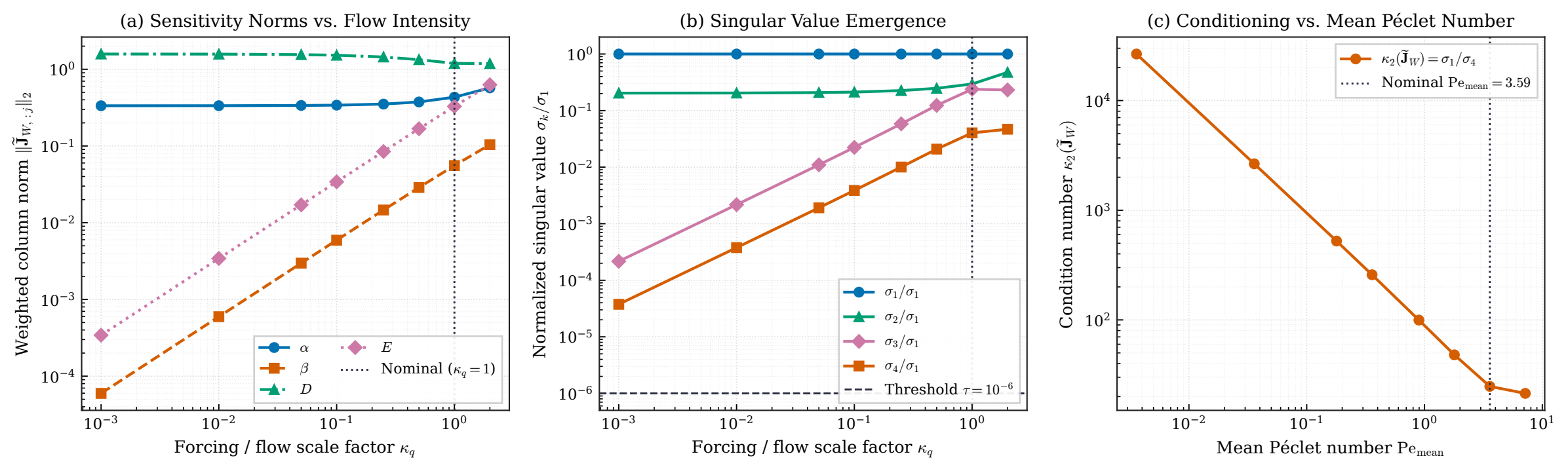
(a) Sensitivity Norms vs. Flow Intensity
Weighted column norm $\|\tilde{\mathbf{J}}_{W,:,j}\|_2$
Forcing / flow scale factor κ_q
α
β
D
E
Nominal ($\kappa_q = 1$)
(b) Singular Value Emergence
Normalized singular value σ_k/σ_1
Forcing / flow scale factor κ_q
σ_1/σ_1
σ_2/σ_1
σ_3/σ_1
σ_4/σ_1
Threshold $\tau = 10^{-6}$
(c) Conditioning vs. Mean Péclet Number
Condition number $\kappa_2(\tilde{\mathbf{J}}_W)$
Mean Péclet number $\mathrm{Pe}_{\mathrm{mean}}$
$\kappa_2(\tilde{\mathbf{J}}_W) = \sigma_1/\sigma_4$
Nominal $\mathrm{Pe}_{\mathrm{mean}} = 3.59$